

Jackie Gao

Multi-Disciplinary Mechanical
Engineer

Timeline



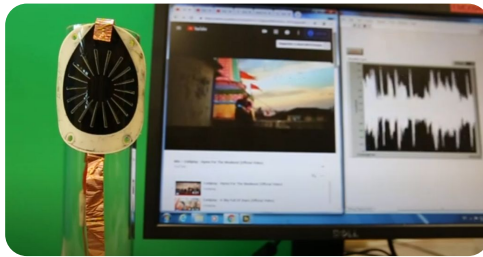
2013 - 2018
B Sc Mechanical
Engineering

2015 - 2018
Airframe Member
Systems Integration Lead
/ President

2018
Student Researcher

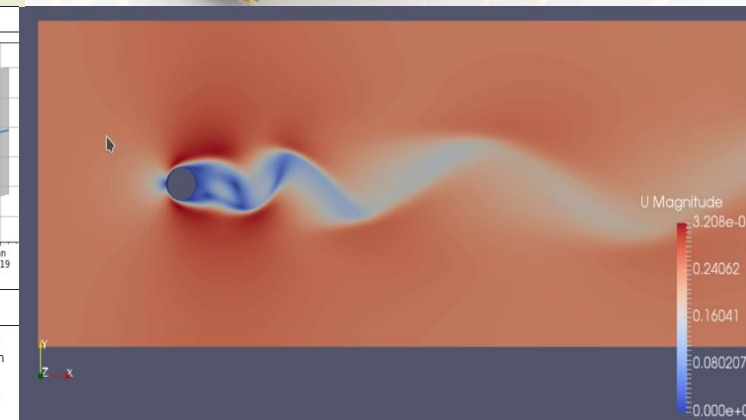
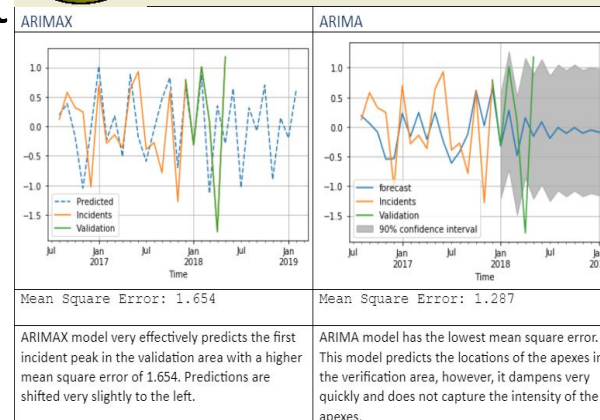
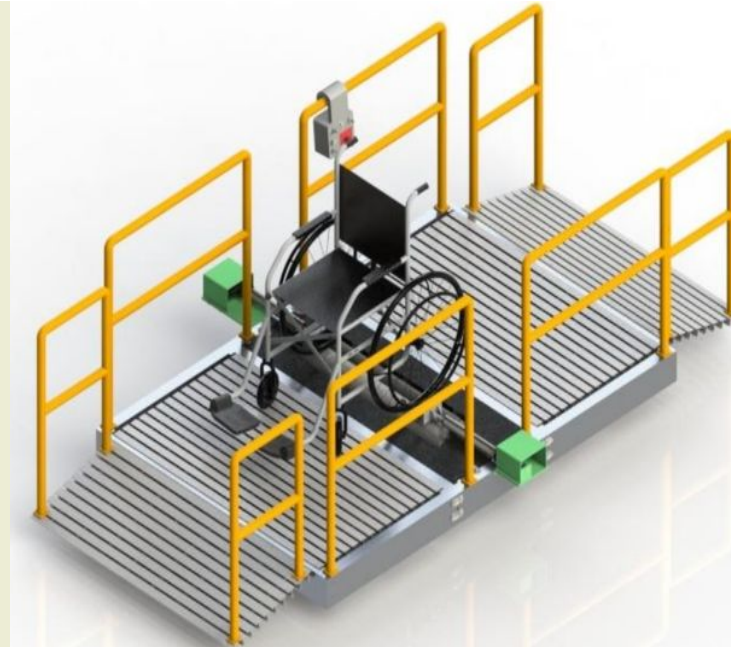
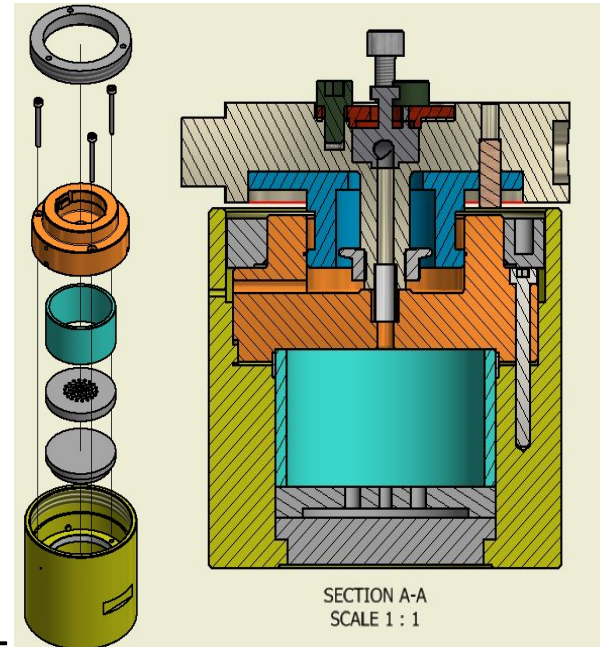
2021 - 2022
Sustaining Engineer

2020 - 2021,
R&D & Lab Automation
2023 - 2025,
Microfluidics & Systems and
Aux. Systems Improvement



University of Alberta

- Academic Projects
 - Capstone: Wheelchair Cleaner
 - Computational Fluid Dynamics
- Shell Canada
 - Quality, Inspection, Email Sorter Macro
- RWTH Textile Institute
 - CAD Redesign of Polymer Extrusion Unit
- Syncrude Canada
 - Project Engineering
- UTSA Adv Manufacturing Lab
 - CNC, Statistical Data Analysis





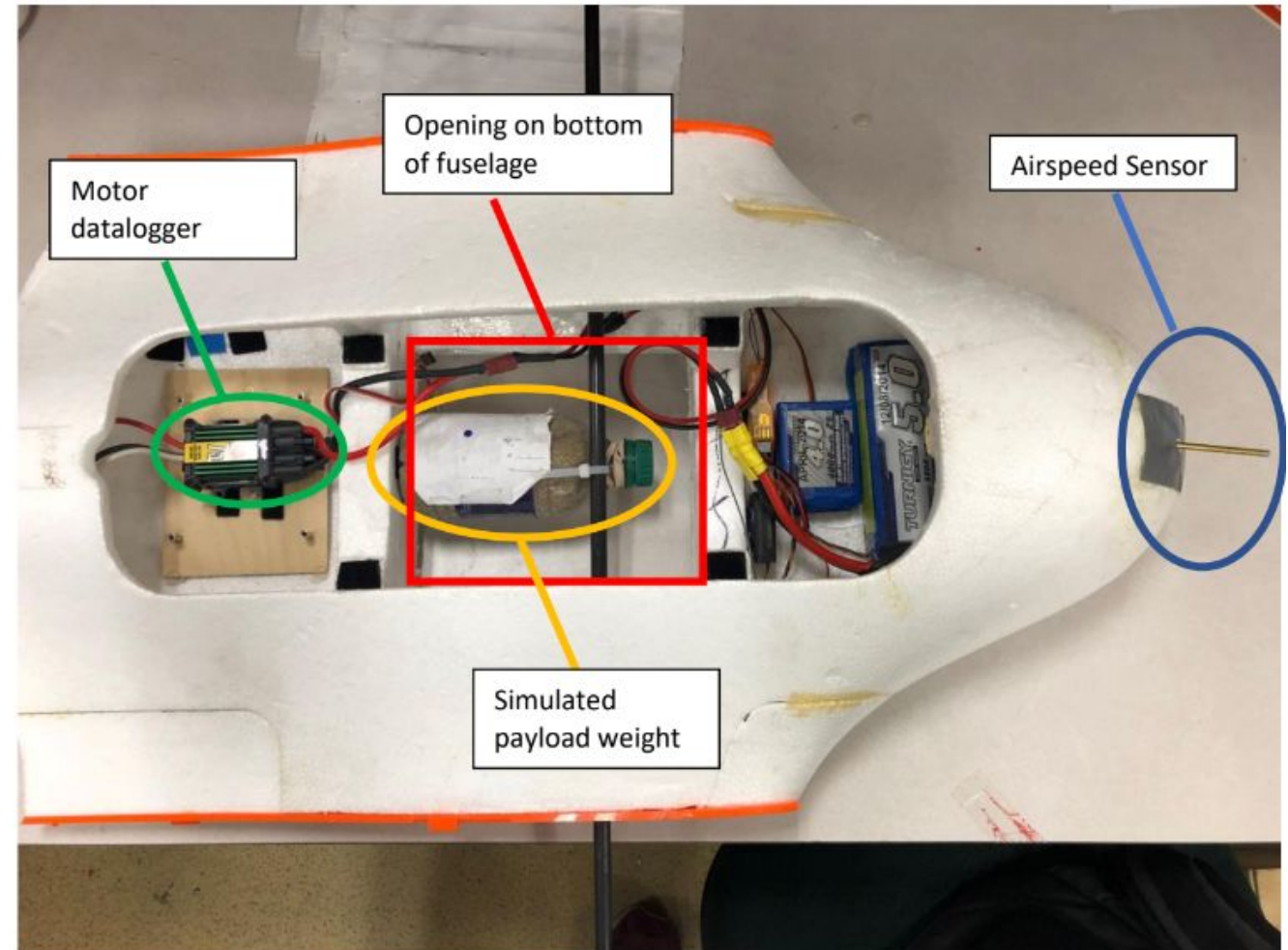
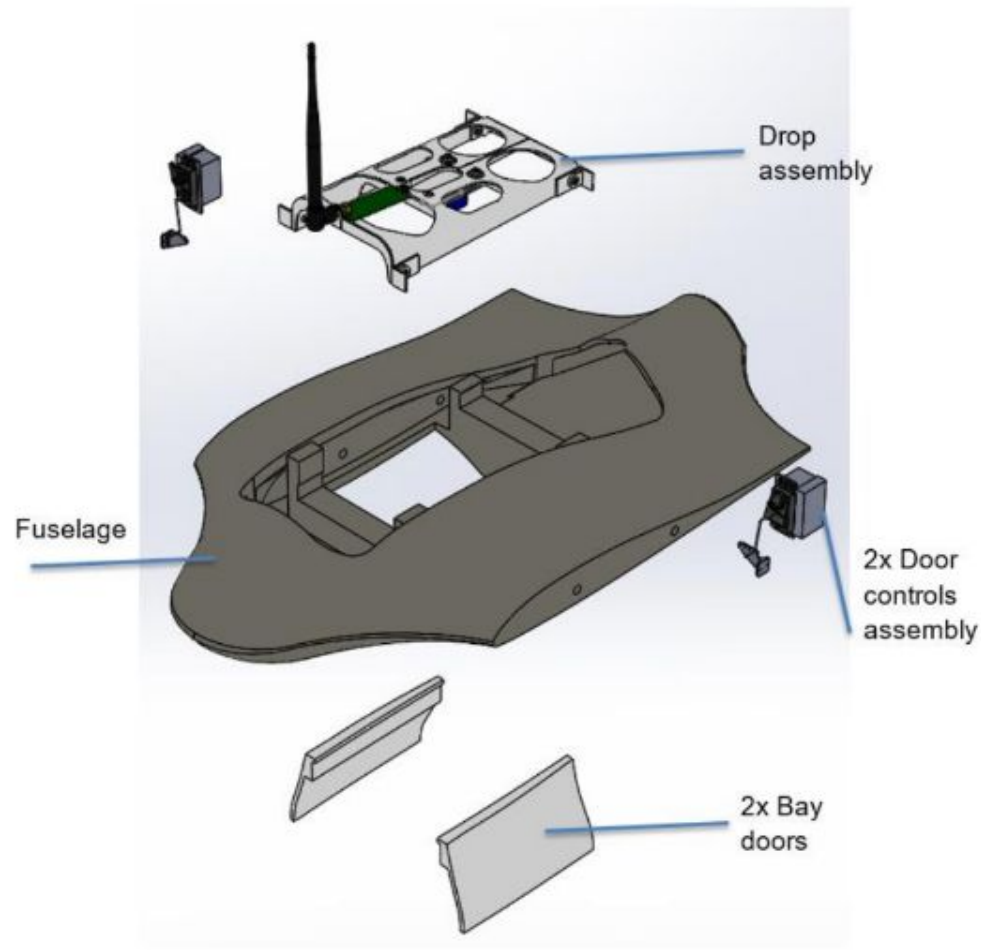
University of Alberta Aerial Robotics Group

- I led a multidisciplinary team of 30 students developing an autonomous UAS for aerial imaging
- I rebuilt this team after the old team graduated
- We flew at AUVSI SUAS 2018 in our first year
- [Paper](#)





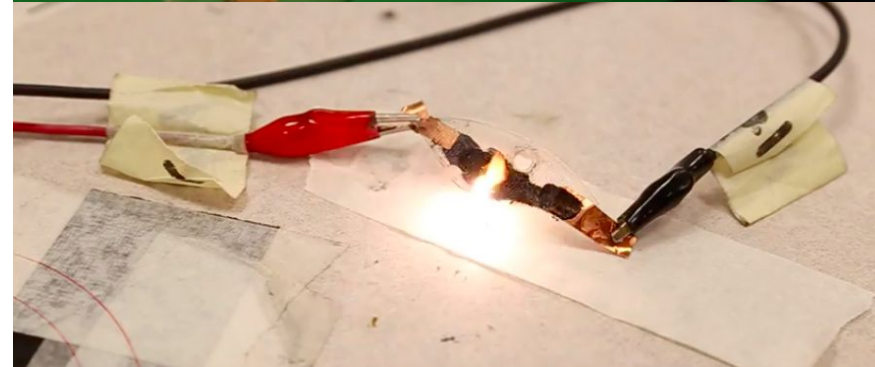
UAARG – Experiment Design for UAV Drop



	Hole Covered	Hole Exposed
Scaled Maximum Lift Coefficient, $C_{l,max}$	1.2	0.66

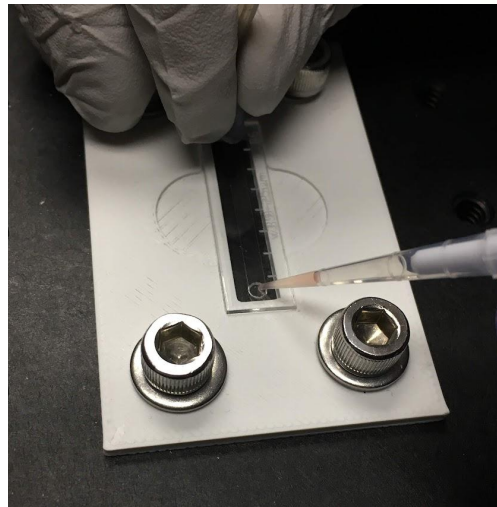
National University of Singapore Soft Robotics

- I traveled to Singapore for more research experience
- I built the **world's first 3D-printed flexible electrostatic loudspeakers**
- Researchers in Singapore are continuing to drive this forward



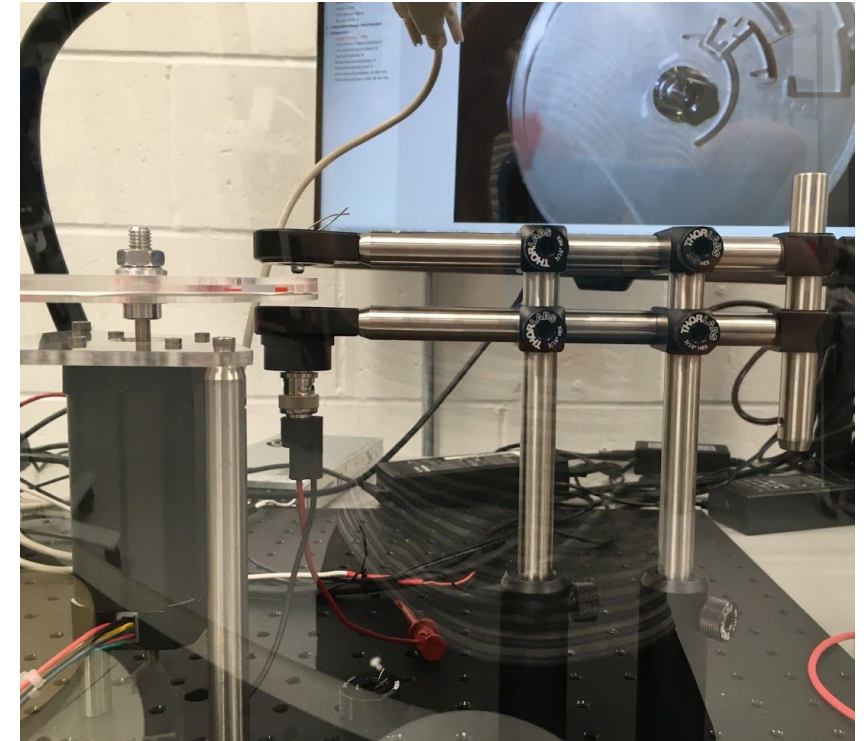
Vital Biosciences – Lab and Liquid Handler Hardware

- Designed and built custom magnetic plate improved the assay from 70% to 90% CV enabling assay studies to be automated
- Built piezoelectric sonicator probe jig for magnetic bead dissociation Enabling bench assay workflow
- Fabricated microfluidic flow cells for complete blood count



Vital Biosciences – System Hardware Design

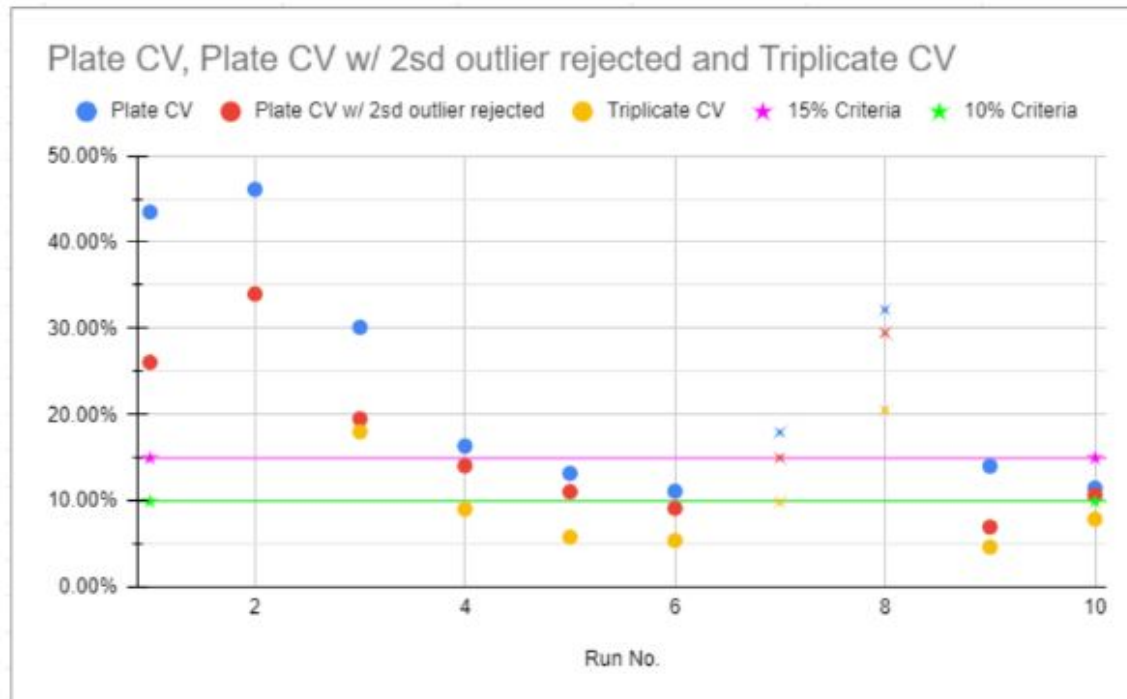
- Designed and built stroboscopic disc monitoring system
- Designed and built a motor system to operate centrifugal microfluidics cartridges
- Designed and built a spectrometry readout system
- Setup data acquisition system with the support of sr software engineer



- Passed Series-A technical assessment (though, funding was not received due to non-technical assessments)

Vital Biosciences – Assay Automation

- Optimized assay from 60% to 90% accuracy
- Enabled assay automation greatly accelerating tech development



Formlabs – Tank Wrinkling Root Causing

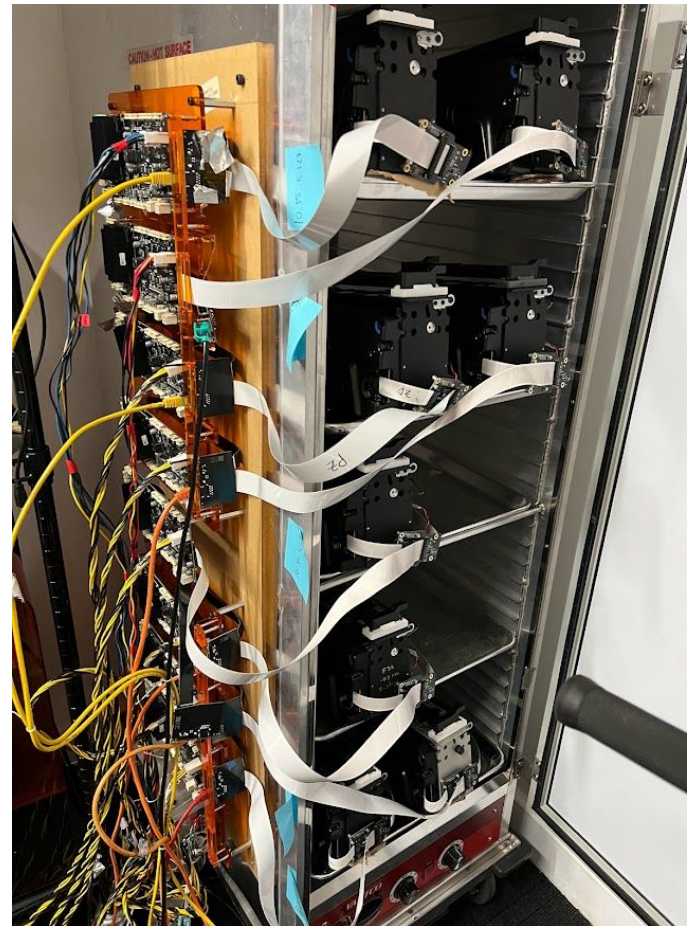
- I moved to Boston to pursue my passion for 3D printing
- Stepped up to tackle Formlab's most elusive and damaging problem
- Tried to understand polymer wrinkling mechanism under tension
- Applied pressure for solving this problem, causing momentum to build
- A solution was implemented, but not mine



frequency:		SME:
SME:		exper
experiment(s):		result
result(s):		
Cure Plane Temperature Only		Cure
hypothesis:	High temperatures at cureplane alone cause warping and delamination of parts	hypot
correctness of hypothesis:	incorrect: high temperatures on tank film results in wrinkling without printing, but does not consistently result in print failures	corre hypot
confidence:	High (experimental backing - Ben Wax)	confi
occurrence frequency:	Assumed low (not many people like to print beefy parts because it is expensive - Pat) - should confirm in data	occu frequ
SME:	Ben Wax	SME
experiment(s):	1. heat 3 Horizontal Bellows tanks and 4 Rev8A tanks and scan for wrinkles 2. print test comparison between Horizontal Bellows tanks and Rev8A tanks	expe
result(s):	1. both HB and Rev8A tanks have demonstrated wrinkles (from flat) when heated to 35C without printing. 2. both Rev 8As and HBs wrinkled under heat but the Rev 8As printed poorly, while HBs printed well. 3. Max temp generated with Black resin is 41°C vs. 53°C generated in Draft, but no significantly higher failure rate in Draft	resu

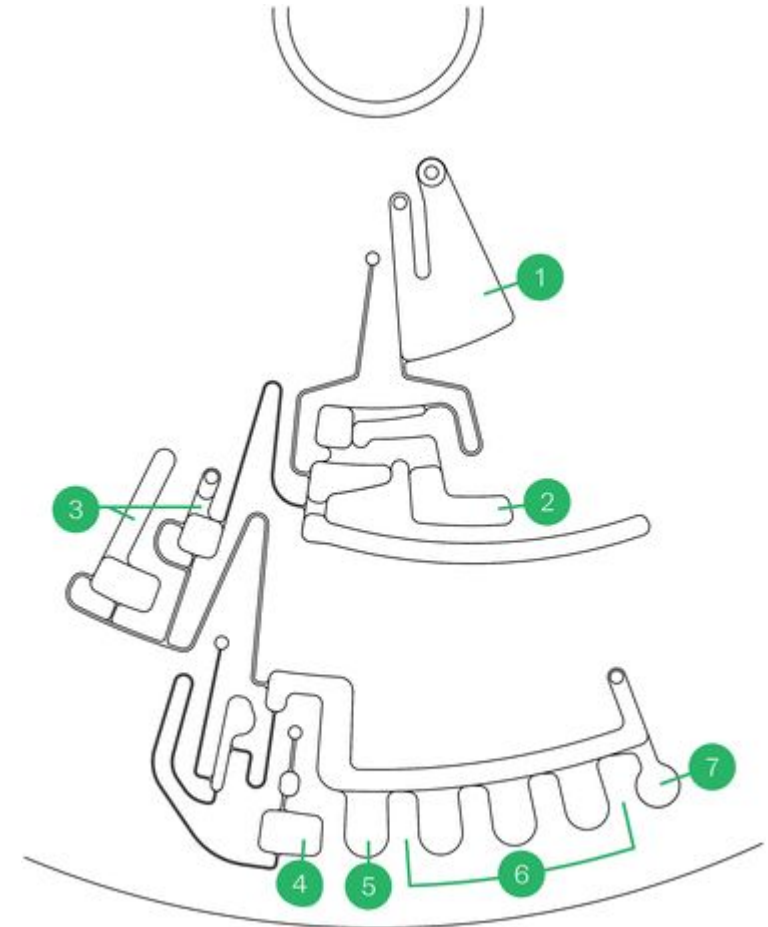
Formlabs – Component Lifetime Testing

- Designed and built lifetime test jigs to validate alternately sourced components
- Characterized component failure modes



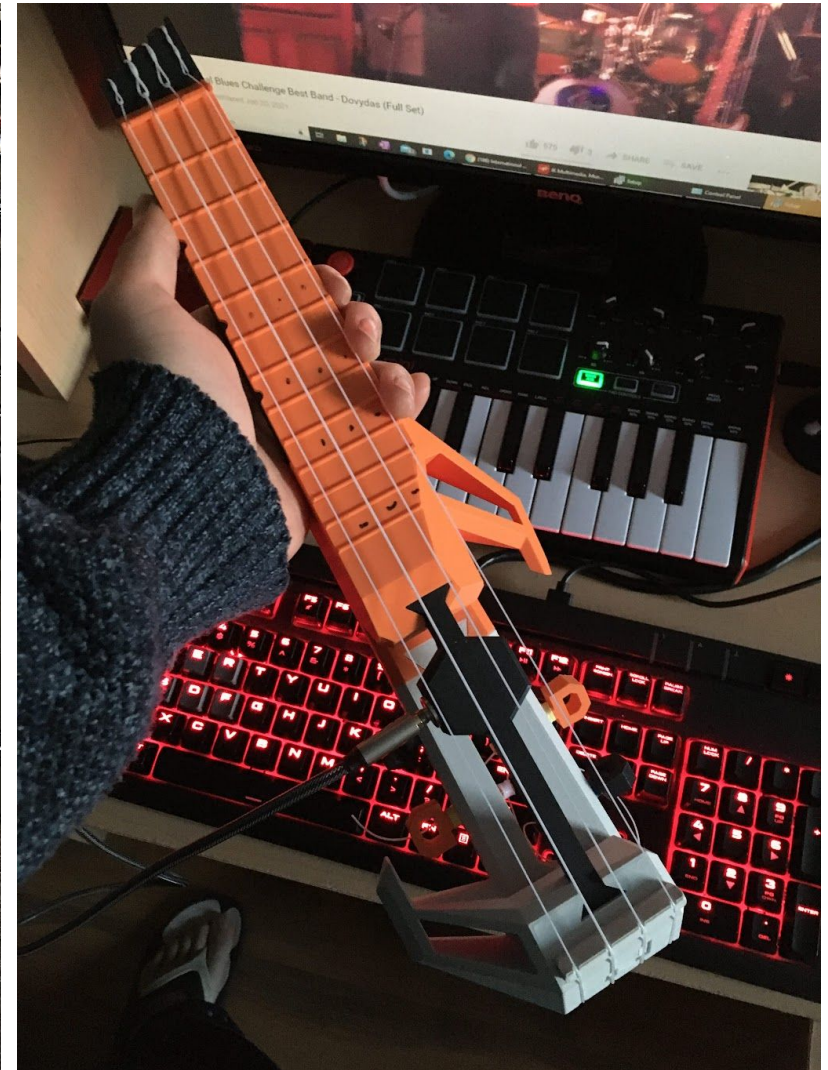
Vital Biosciences - Microfluidics, Systems Improvement and Auxiliary Systems

- Assessed system thermal conditions and root caused assay performance issues pointing to heat distribution
- Improved microfluidics spinning disc visualization system working with systems engineering
- Troubleshoot and increase reliability of microfluidics systems working with oversea injection molding vendors
- Supporting across the company, root causing and rapid prototyping efforts
- Though I'm not listed as an author, [this paper overviews the system I worked on with Vital Biosciences](#)



Personal – Electric Ukulele

Piezoelectric
electric-ukulele



Hobbies

